

Overview and Analysis of Current Sources on the Euphemia Algorithm: Foundation for Implementation in the Course of the APEM Project

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1 Introduction

In the following, a summary of the available sources on the Euphemia algorithm will be provided, which will serve as a foundation for an implementation for the APEM-Project.

It should be mentioned in advance that there are two publicly accessible implementations. The first implementation is based on the paper [3] and is freely available on [GitHub](#) under a [BSD 2-Clause License](#). The outlined algorithm focuses on curtailable block orders and the uniform purchase price (PUN orders). Since we do not focus on PUN orders for our implementation, this implementation will not be used for the time being. A second publicly accessible implementation is based on a simplification of the algorithm in Chapter 4 from [1]. For easier understanding, the algorithm is presented in Chapter 3 and can be found on [GitHub](#). Since the algorithm has been simplified to a single bidding zone and therefore does not consider, among other things, the congestion rent, the implementation is too simplified for our application.

The insights from figure 1 in [4] draw our attention to [1] for our requirements of all orders except for PUN and Merit orders, which we will address in chapter 4.

Article	Stepwise hourly	Interpolated hourly	Block	Block with MAR	Linked blocks	Exclusive blocks	Flexible	Complex	PUN
[31] O'Neill et al. (2005)	✓		✓						
[36] O'Neill et al. (2007)	✓		✓						
[23] Basagoiti et al. (2008)	✓		✓					✓	
[37] Meeus et al. (2009)	✓								
[29] Meeus et al. (2009)	✓		✓						
[12] Van Vyve (2011)	✓		✓						
[33] Biskas et al. (2014)	✓		✓		✓		✓	✓	
[28] Madani and Van Vyve (2014)	✓		✓						
[38] Tjersansen and Rudlang (2014)	✓		✓						
[9] Madani and Van Vyve (2014)	✓		✓					✓	
[4] Martin et al. (2014)	✓	✓	✓		✓		✓		
[25] Madani and Van Vyve (2015)	✓	✓	✓						
[32] Derinkuyu (2015)		✓	✓		✓		✓		
[13] Dourbois and Biskas (2015)	✓		✓		✓	✓	✓	✓	
[10] Chatzigiannis et al. (2016)	✓		✓	✓	✓	✓	✓	✓	
[5] Dourbois and Biskas (2017)	✓		✓				✓	✓	✓
[11] Madani and Van Vyve (2017)	✓		✓					✓	
[39] Madani and Van Vyve (2018)	✓		✓					✓	
[6] Dourbois et al. (2018)	✓		✓				✓	✓	✓
[3] Lam et al. (2018)	✓	✓	✓		✓	✓	✓	✓	✓
[2] Derinkuyu et al. (2019)	✓	✓	✓		✓		✓		

Figure 1: Overview about existing paper works from [4]

2 Welfare Maximization Problem, Euphemia Public Description

The following Objective is the original model from the Euphemia Public Description ([2]).

$$\max \quad - \sum_{m,h,s,o: \text{Step Orders}} \text{Accept}_{m,h,s,o} \cdot q_{m,h,s,o} \cdot p_{m,h,s,o}^0 \quad (1a)$$

$$- \sum_{m,h,s,o: \text{Interpolated Orders}} \text{Accept}_{m,h,s,o} \cdot q_{m,h,s,o} \cdot (p_{m,h,s,o}^0 + \text{Accept}_{m,h,s,o} \cdot \frac{p_{m,h,s,o}^1 - p_{m,h,s,o}^0}{2}) \quad (1b)$$

$$- \sum_{bo,h} \text{Accept}_{bo} \cdot q_{bo,h} \cdot p_{bo} \quad (1c)$$

$$- \sum_{m,co,h} \text{Accept}_{m,co,h,o} \cdot q_{m,co,h,o} \cdot p_{m,co,h,o} \quad (1d)$$

$$- \sum_{mo} \text{Accept}_{mo} \cdot q_{mo} \cdot p_{mo} \quad (1e)$$

$$- \sum_{l,u,h} \text{Tariff}_{l,h} \cdot \text{Flow}_{l,u,h} \quad (1f)$$

$$- M \sum_{m,h,o: \text{PriceTakingHourlyDemandOrders}} \text{Max_Curtailment_Ratio} \quad (1g)$$

Explanation:

(2a) : Welfare contribution of hourly step orders

(2b) : Welfare contribution of hourly interpolated orders

(1c) : Welfare contribution of block orders

(1d) : Welfare contribution of complex orders

(1e) : Welfare contribution of merit orders

(1f) : Welfare impact o tariffs

(1g) : If the value of M is sufficiently large, it will help minimize the rejected price-taking quantity in all markets, before looking for a solution with a good welfare.

Subject to:

- Market constraints:
 - Balance/clearing constraints
 - Block order acceptance constraint
 - Complex suborders acceptance constraints
 - Load Gradient constraint
 - Merit order acceptance constraints
- Network constraints:
 - ATC constraints
 - PDTF constraints
 - Carious ramping constraints

Index	Description
m	Bidding zone
h	Period
s	Supply/Demand
c	Curve identified by m,h,s
o	Hourly Order identified by m,h,s,o
bo	Block Order
mo	Merit order
po	PUN order
co	Complex Order, where · Complex curve is identified by m,co,h · Complex suborder by m,co,h,o
l	(DC/ATC) Line
uu	(convention: up=0 and down=1) Up/Down direction
Accept	[0;1] Acceptance variables
p	Offered Price
q	Offered Volume
MCP	Market clearing price

3 Simplified Clearing Model for the Day-Ahead Market

3.1 Objective Function

$$\max_{x,u} w_{\text{tot}} = ut_{\text{dem}} - c_{\text{sup}} + w_{\text{block}} \quad (2a)$$

$$= \sum_{d \in D} x_d \cdot P_d \cdot Q_d - \sum_{s \in S} x_s \cdot P_s \cdot Q_s + \sum_{b \in B} \sum_{t \in T} x_b \cdot P_b \cdot Q_b^t \quad (2b)$$

(w_{tot}) : Social Welfare without congestion rent, because in the simplification exists only one bidding zone

(ut_{dem}) : Utility of demand side, excluding block orders

(c_{sup}) : Suppliers' cost, excluding block orders

(w_{block}) : Contribution of block orders to social welfare

Variables : x: acceptance ratio

Parameter : P: price; Q: quantity

Index sets : D: hourly demand orders; S: hourly supply orders; B: block orders; T: trading periods within a day (typically an hour)

Subject to:

3.2 Order Clearing Constraints

$$x_d \leq 1, \quad \forall d \in D \quad (3)$$

$$x_s \leq 1, \quad \forall s \in S \quad (4)$$

$$x_b \leq u_b, \quad \forall b \in B \quad (5)$$

$$x_b \geq R_b \cdot u_b \quad \forall b \in B \quad (6)$$

$$x_{lb} \leq \sum_{b \in B} A_b^{lb} \cdot x_b, \quad \forall lb \in LB \quad (7)$$

$$\sum_{b \in B} A_b^{eg} \cdot x_b \leq 1, \quad \forall eg \in EG \quad (8)$$

$$\sum_{t \in T} u_{fh}^t \leq 1, \quad \forall fh \in FH \quad (9)$$

(3) + (4) : Upper limit for the acceptance values of hourly demand + supply orders

(5) + (6) : Lower (minimum acceptance ratio (R) * Acceptance Status (u (binary))) + upper (if fully rejected $u = 0$ otherwise $u = 1$) bound for block order acceptance value

(7) : In linked block orders, the acceptance ratio of a child block can not be larger than the one of its parent block

(8) : Combined acceptance ratio of all blocks contained in an exclusive group may not exceed 1

(9) : Binary acceptance u_{fh}^t can only take 1 for exactly one $t \in T$

Index sets : LB: linked block order; EG: exclusive group of block orders; FH: flexible hourly orders

Parameter : A_b^{lb} : 1 if block order lb is linked child of block b , else 0; A_b^{eg} : 1 if block b is part of exclusive group eg , else 0

3.3 Power Balance Constraints

$$v_{\text{demand}}^t = \sum_{d \in D} x_d \cdot Y_d^t \cdot Q_d \quad \forall t \in T \quad (10)$$

$$v_{\text{supply}}^t = \sum_{s \in S} x_s \cdot Y_s^t \cdot Q_s \quad \forall t \in T \quad (11)$$

$$v_{\text{block}}^t = \sum_{b \in B} x_b \cdot Q_b^t \quad \forall t \in T \quad (12)$$

$$v_{\text{demand}}^t + v_{\text{block}}^t = v_{\text{supply}}^t \quad \forall t \in T \quad (13)$$

(10) + (11) : Volume for a period of supply + demand of the hourly orders equals the sum of all accepted orders of the respective side in that period. The binary viable Y_i^t denotes whether the order $i \in \{d, s\}$ was entered for period t

(12) : Du to supply block orders being modeled with negativ equantities, the volume of block orders in a period already covers demand, as well as supply that is cleared in the specific hour of all accepted block orders

(13) : Cleared energy on the demand side is matched by the cleared supply

4 European day-ahead electricity market clearing model, [1]

4.1 Input

- Orders (specific definition in Chapter 5):
 - Aggregated hourly order (Step-wise linear hourly supply and demand orders)
 - Complex orders: supply offers with special clearing rules (Load Gradient Condition, Minimum Income Condition, Scheduled Stop Condition)
 - Block orders: regular block orders, profile block orders (allowing the partial acceptance of the order) linked block orders (acceptance ratio of parent block should always be greater than that of the child), block orders in an exclusive group and flexible hourly orders
- Not available (compared to [2]): PUN/ Merit orders, aggregated piecewise hourly orders, scalable complex orders
- Network Modeling: Flow Based Model (PTDF)

4.2 Solution Process

- Initially, the overall MILP problem is solved and the optimally cleared volumes of all orders are attained
- The MCPs for each bidding area and trading period are computed
- Algorithm performs two consecutive checks: Paradoxically Accepted Orders (PAOs) and Minimum Income Condition (MIC)
- If one is not satisfied, then the algorithm continues with step (a).

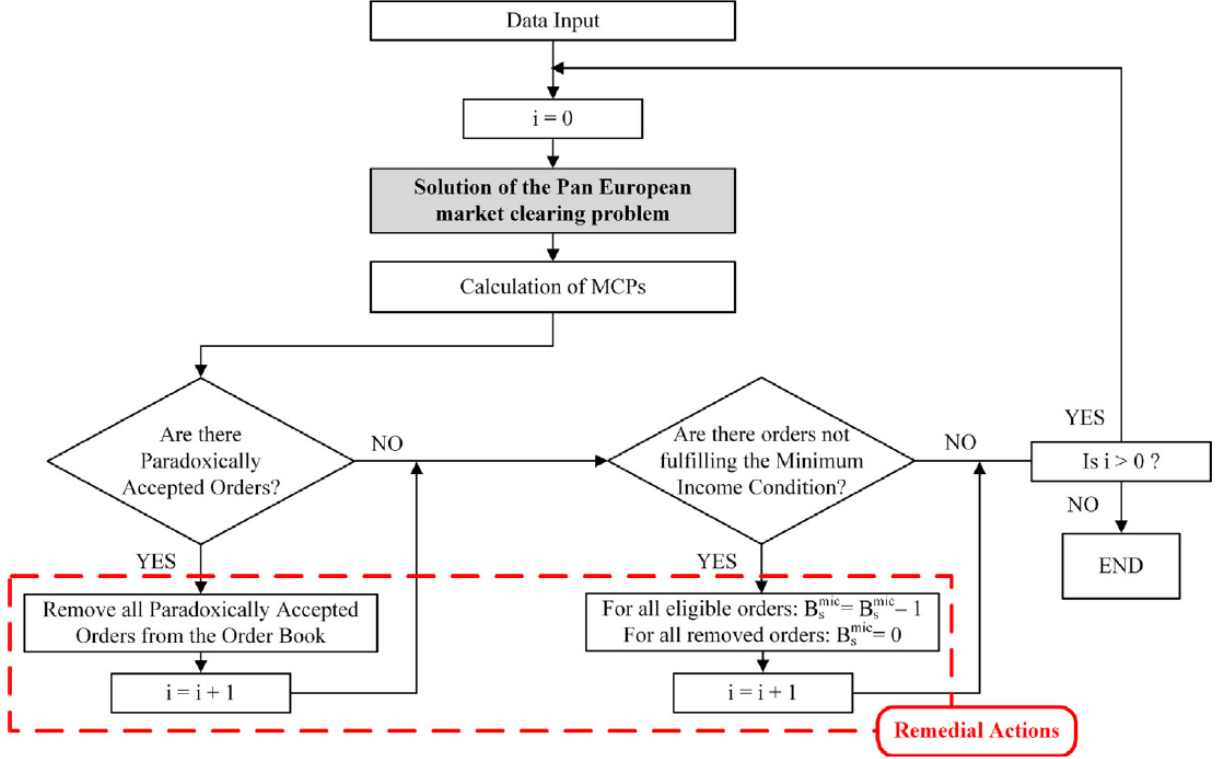


Figure 2: Schematic illustration of the iterative algorithm

4.3 MILP model

4.3.1 Objective Function

$$\max_{v \in V} w_{tot} = u_{dem} - (c_{sup} + c_{bo} + c_{fho}) - c_{cr}, \text{ with} \quad (14)$$

$$u_{dem} = \sum_{d \in D} \sum_{b \in B} \sum_{t \in T} (P_{db}^t \cdot Q_{db}^t \cdot x_{db}^t) \quad (15)$$

$$c_{sup} = \sum_{s \in S} \sum_{b \in B} \sum_{t \in T} (P_{sb}^t \cdot Q_{sb}^t \cdot x_{sb}^t) \quad (16)$$

$$c_{bo} = \sum_{bo \in BO} \sum_{t \in T} (P_{bo} \cdot Q_{bo}^t \cdot x_{bo}) \quad (17)$$

$$c_{fho} = \sum_{fho \in FHO} \sum_{t \in T} (P_{fho} \cdot Q_{fho} \cdot u_{fho}^t) \quad (18)$$

$$c_{cr} = \sum_{l \in L_{dc}} \sum_{t \in T} (\tau_{lt} \cdot (f_{lt}^+ + f_{lt}^-)) \quad (19)$$

(w_{tot}) : Objective function (maximization of social welfare) of the MILP optimization problem under Set of primal decision variables $V = \{p_{at}, f_{lt}, e_{aa'}^t, x_{sb}^t, x_{db}^t, x_{bo}, u_{bo}, u_{fho}^t\}$

(u_{dem}) : the total load utility function (Consumer surplus: Accepted Price-Quantity pair of the demand offers)

(c_{sup}) : total simple hourly offer cost function (producer surplus: Accepted Price-Quantity pair of the supplied offers)

- (c_{bo}) : total block offer cost function (producer surplus: Accepted Price-Quantity pair of the supplied offers)
- (c_{fho}) : total flexible hourly offer cost function (producer surplus: Accepted Price-Quantity pair of the supplied offers)
- (c_{cr}) : congestion rent (related tariffs of DC interconnections)

Dec. Variables	Description
p_{at}	net position of bidding area a in trading period t , in MWh; the term “net position” refers to the net injection of a bidding area, namely total cleared production minus total cleared demand.
f_{lt}	power flow in interconnection l in trading period t , in MW
$e_{aa'}^t$	bilateral exchange between bidding area a and bidding area a' , in trading period t , in MWh
$x_{sb}^t(x_{db}^t)$	acceptance ratio of step b of supply (demand) entity's s priced offer (bid) in trading period t
x_{bo}	acceptance ratio of block order bo
u_{bo}	clearing status of block order bo (binary)
u_{fho}^t	clearing status of flexible hourly order fho in trading period t (binary)

4.3.2 Constraints

Order clearing constraints:

$$x_{sb}^t \leq 1 \quad \forall s \in S, b \in B, t \in T \quad (20)$$

$$x_{db}^t \leq 1 \quad \forall d \in D, b \in B, t \in T \quad (21)$$

$$\sum_{b \in B} Q_{sb}^t x_{sb}^t - \sum_{b \in B} Q_{sb}^{t-1} x_{sb}^{t-1} \leq 60 G_{up}^s \quad \forall s \in S_{lg}, t \in T - \{t1\} \quad (22)$$

$$\sum_{b \in B} Q_{sb}^{t-1} x_{sb}^{t-1} - \sum_{b \in B} Q_{sb}^t x_{sb}^t \leq 60 G_{dn}^s \quad \forall s \in S_{lg}, t \in T - \{t1\} \quad (23)$$

$$x_{sb}^t \leq B_{mic}^s + B_{ss}^{sbt} \quad \forall s \in S_{mic}, b \in B, t \in T \quad (24)$$

$$R_{min}^{bo} u_{bo} \leq x_{bo} \quad \forall bo \in BO \quad (25)$$

$$x_{bo} \leq u_{bo} \quad \forall bo \in BO \quad (26)$$

$$x_{lbo} \leq \sum_{bo \in BO} A_{lbo}^{bo} x_{bo} \quad \forall lbo \in LBO \quad (27)$$

$$\sum_{bo \in BO} A_{eg}^{bo} x_{bo} \leq 1 \quad \forall eg \in EG \quad (28)$$

$$\sum_{t \in T} u_{fho}^t \leq 1 \quad \forall fho \in FHO \quad (29)$$

(20) + (21) : upper limits for the clearing of the simple hourly supply offers and demand

(22) + (23) : enforce the Load Gradient Condition to the respective supply orders (Faktor 60 changes the unit of $G_{up/dn}^s$ from MW/min to MW/s.)

(24) : enforces simultaneously the Minimum Income and Scheduled Stop Conditions

(25) + (26) : enforce the clearing status of a profile block order should always be zero or btw. its min and max acceptance ratio

(27) : acceptance condition of a linked block order lbo

(28) : sum of the accepted ratios of block orders belonging to exclusive group eg to be less than or equal to 1

(29) : clearing condition of flexible hourly orders (only one acceptance per order)

Power balance constraints:

$$p_{at} = \sum_{s \in S_a} \sum_{b \in B} Q_{sb}^t x_{sb}^t - \sum_{d \in D_a} \sum_{b \in B} Q_{db}^t x_{db}^t + \sum_{bo \in BO_a} Q_{bo}^t x_{bo}^t + \sum_{fho \in FHO_a} Q_{fho}^t u_{fho}^t \quad \forall a \in A, t \in T \quad [\xi_{at}] \quad (30)$$

$$p_{at} = \sum_{a' \in A} (e_{aa'}^t - e_{a'a}^t) + \sum_{l \in L_{dc}} (\chi_l^a (f_{lt}^+ - (1 - \lambda_l^a) f_{lt}^-)) - \sum_{l \in L_{dc}} (\varphi_l^a ((1 - \lambda_l^a) f_{lt}^+ - f_{lt}^-)) \quad \forall a \in A, t \in T \quad (31)$$

(30) : provides the net position of bidding area a in trading period t (Sum of all accepted demand and supply order per area a)

(31) : expresses the hourly power balance equation of each bidding area a (Sum of bilateral exchange minus losses if there is an DC interconnector involved)

Power flow constraints:

$$f_{lt} = \sum_{a \in A} \sum_{a' \in A} PTDF_{aa'}^{lt} \cdot e_{aa'}^t \quad \forall l \in L_{AC}, t \in T \quad (32)$$

$$f_{lt} = f_{lt}^+ - f_{lt}^- \quad \forall l \in L, t \in T \quad (33)$$

$$f_{lt} \leq F_{max}^{lt} \quad \forall l \in L, t \in T \quad (34)$$

$$F_{min}^{lt} \leq f_{lt} \quad \forall l \in L, t \in T \quad (35)$$

(32) - (35) : The opt. model incorporates a DC Transmission network configuration (without AC Power flow constraints), where (32) is used for the modeling of the power flows of the AC interconnectors

Hourly and daily net position ramping constraints:

$$p_{at} - p_{a,t-1} - P_{at}^{H,ini} \leq R_{at}^{H,up} \quad \forall a \in A, t \in T \quad [\beta_{at}^{H,up}] \quad (36)$$

$$p_{a,t-1} - p_{at} + P_{at}^{H,ini} \leq R_{at}^{H,dn} \quad \forall a \in A, t \in T \quad [\beta_{at}^{H,dn}] \quad (37)$$

$$\sum_{t \in T} p_{at} - P_a^{D,ini} \leq R_a^{D,up} \quad \forall a \in A \quad [\beta_a^{D,up}] \quad (38)$$

$$P_a^{D,ini} - \sum_{t \in T} p_{at} \leq R_a^{D,dn} \quad \forall a \in A \quad [\beta_a^{D,dn}] \quad (39)$$

(36) - (37) : hourly net position ramping constraint

(38) - (39) : daily net position ramping constraint

Hourly individual line and line set power flow ramping constraints:

$$f_{lt} - f_{l,t-1} - F_{lt}^{H,ini} \leq R_{lt}^{H,up} \quad \forall l \in L, t \in T \quad (40)$$

$$f_{l,t-1} - f_{lt} + F_{lt}^{H,ini} \leq R_{lt}^{H,dn} \quad \forall l \in L, t \in T \quad (41)$$

$$\sum_{l \in L_{ls}} f_{lt} - \sum_{l \in L_{ls}} f_{l,t-1} - F_{ls,t}^{LS,ini} \leq R_{ls}^{LS,up} \quad \forall ls \in LS, t \in T \quad (42)$$

$$\sum_{l \in L_{ls}} f_{l,t-1} - \sum_{l \in L_{ls}} f_{lt} + F_{ls,t}^{LS,ini} \leq R_{ls}^{LS,dn} \quad \forall ls \in LS, t \in T \quad (43)$$

(40) - (41) : hourly flow ramping constraints on individual lines; analogous to the hourly net position ramping constraints

(42) - (43) : hourly flow ramping constraints on line sets

4.3.3 Market Clearing Prices

$$\pi_{at} = \zeta_{at} - (\beta_{H,up}^{at} - \beta_{H,up}^{a,t+1}) + (\beta_{H,dn}^{at} - \beta_{H,dn}^{a,t+1}) - \beta_{D,up}^a + \beta_{D,dn}^a \quad \forall a \in A, t \in T \quad (44)$$

4.3.4 Paradoxically accepted orders check

$$W_{bo} = \sum_{t \in T} (\pi_{at} - P_{bo}) \cdot Q_{bo}^t \cdot \bar{x}_{bo} \quad \forall bo \in BO_a \quad (45)$$

$$W_{fho} = \sum_{t \in T} (\pi_{at} - P_{fho}) \cdot Q_{fho} \cdot \bar{u}_{bo}^t \quad \forall fho \in FHO_a \quad (46)$$

If welfare (W_{bo}, W_{fho}) is negative, then the respective Paradoxically accepted block and flexible hourly order are effectively removed from the order book. $\bar{x}_{bo}$ and $\bar{u}_{bo}^t$ are the optimal values of x_{bo} and u_{bo}^t .

4.4 Nomenclature

Sets	Description
$t(T)$	index (set) of trading periods within the trading day (typically, the period is 1 h)
$a(A)$	index (set) of bidding areas
$b(B)$	index (set) of supply offers' and demand bids' steps
$d(D)$	index (set) of demand bids, $D_a \subseteq D$ is the subset of demand bids submitted to bidding area a
$eg(EG)$	index (set) of exclusive groups
$l(L)$	index (set) of interconnections, where $L_{ac} \cup L_{dc} = L$ are the subsets of AC and DC interconnections and $L_{ls} \subseteq L$ is the subset of interconnections belonging to the same line set
$s(S)$	index (set) of supply, $S_a \subseteq S$ is the subset of supply offers submitted to bidding area a , $S_{lg} \subseteq S$ the subset of supply offers that are subject to a Load Gradient Condition, $S_{mic} \subseteq S$ the subset of supply offers that are subject to a Minimum Income Condition and $S_{ss} \subseteq S$ the subset of supply offers that are subject to a Scheduled Stop Condition
$bo(BO)$	index (set) of block orders (profile or regular block orders), $BO_a \subseteq BO$ is the subset of block orders submitted to bidding area a
$lbo(LBO)$	index (set) of linked block orders, $LBO_a \subseteq LBO$ is the subset of linked block orders submitted to bidding area a
$fho(FHO)$	index (set) of flexible hourly orders, $FHO_a \subseteq FHO$ is the subset of flexible hourly orders submitted to bidding area a
$ls(LS)$	index (set) of line sets

Parameters	Source	Description
P_{db}^t, Q_{db}^t	Orders	price-quantity pair of step b of the hourly priced demand bid d in trading period t , in €/MWh and MWh, respectively
P_{sb}^t, Q_{sb}^t	Orders	price-quantity pair of step b of the hourly priced energy offer s in trading period t , in €/MWh and MWh, respectively
P_{bo}, Q_{bo}^t	Orders	price-quantity pair of block order bo , in €/MWh and MWh, respectively; in the case of a profile block order, the quantity may be different in each trading period t
P_{fho}, Q_{fho}	Orders	price-quantity pair of flexible hourly order fho , in €/MWh and MWh, respectively
$F_{lt}^{min}(F_{lt}^{max})$	Estimate	minimum (maximum) available capacity (also called Remaining Available Margin in the flow-based model) of interconnection l in trading period t , in MW
R_{bo}^{min}	Orders (BO)	Minimum Acceptance Ratio of block order bo , in p.u.
$PTDF_{aa'}^{lt}$	From OPF (PyPSA)	power transfer distribution factor (PTDF) of AC interconnection l for an energy transfer (exchange) between bidding areas a and a' in trading period t , in p.u.
A_{bo}^{lbo}	Orders (linked BO)	linked block order lbo to block order bo incidence matrix
A_{bo}^{eg}	Orders (excl. BO)	block order bo to exclusive group eg incidence matrix
τ^{lt}	Estimate	flow tariff for the utilization of DC interconnection l in trading period t , in €/MW h

Parameters	Source	Description
$G_s^{dn}(G_s^{up})$	set as a constant (Estimate)	decrease (increase) gradient of the supply order s , which is subject to the Load Gradient Condition in MW/min
B_{sbt}^{ss}	Orders (Stop Schedule)	ancillary parameter denoting the activation of the Scheduled Stop Condition, if equal to 1
B_s^{mic}	Orders (MIC Option)	ancillary parameter denoting the number of available opportunities a supply offer s has, in order to fulfil its Minimum Income Condition
$\chi_l^a(\varphi_l^a)$	Net Info	parameter denoting that DC interconnection l begins (ends) from (to) bidding area a , if equal to 1; otherwise it is equal to 0
λ_l^a	Estimate	loss factor of DC interconnection l when power is transferred to bidding area a , in %
$R_{at}^{H,dn}(R_{at}^{H,up})$	Not shure Estimate?	hourly ramp-down (ramp-up) limit of bidding area's a net position, in trading period t , in MW/h; the term "net position ramping" refers to the hourly difference in the net position of a bidding area between two consecutive trading periods of the trading day.
$P_{at}^{H,ini}$	Results of preday $= p_{a(t-24-(tmod24))}$	parameter denoting the initial net position of bidding area a in trading period t ; it is equal to the net position of bidding area a during the last hour of the previous trading day for the first trading period ($t = 1$) and zero for all other trading periods ($t \in \{2 - 24\}$), in MW
$R_a^{D,dn}(R_a^{D,up})$	Not shure Estimate?	daily ramp-down (ramp-up) limit of bidding area's a net position, in MW (per day)
$P_a^{D,ini}$	Results of preday	total net position of bidding area a during the previous trading day, in MW
$R_{lt}^{H,dn}(R_{lt}^{H,up})$	Not shure Estimate?	hourly ramp-down (ramp-up) limit of interconnection l power flow, in trading period t , in MW/h
$F_{lt}^{H,ini}$	Results of preday $= p_{a(t-24-(tmod24))}$	parameter denoting the initial power flow in interconnection l in trading period t ; it is equal to the power flow of interconnection l during the last hour of the previous trading day for the first trading period ($t = 1$), and zero for all other trading periods ($t \in \{2 - 24\}$), in MW
$R_{ls}^{LS,dn}(R_{ls}^{LS,up})$	Not shure	ramp-down (ramp-up) limit of line set ls total power flow, in MW/h
$F_{ls,t}^{LS,ini}$	Results of preday	parameter denoting the initial total power flow in line set ls in trading period t ; it is equal to the total power flow of line set ls during the last hour of the previous trading day for the first trading period ($t = 1$) and zero for all other trading periods ($t \in \{2 - 24\}$), in MW

Positive Variables	Description
p_{at}	net position of bidding area a in trading period t , in MWh; the term “net position” refers to the net injection of a bidding area, namely total cleared production minus total cleared demand.
f_{lt}	power flow in interconnection l in trading period t , in MW
$e_{aa'}^t$	bilateral exchange between bidding area a and bidding area a' , in trading period t , in MWh
$x_{sb}^t (x_{db}^t)$	acceptance ratio of step b of supply (demand) entity's s priced offer (bid) in trading period t
x_{bo}	acceptance ratio of block order bo
f_{lt}^+, f_{lt}^-	positive variables utilized in the power flow variable decomposition schema for DC interconnection l in trading period t , in MW
Binary Variables	
u_{bo}	clearing status of block order bo
u_{fho}^t	clearing status of flexible hourly order fho in trading period t

5 Orders

Compare with [2].

5.1 Aggregated Hourly Orders

Demand (resp. supply) orders from all market participants belonging to the same bidding zone will be aggregated into a single curve referred to as aggregated demand (resp. supply) curve defined for each period of the day. Demand orders are sorted from the highest price to the lowest. Conversely, supply orders are sorted from the lowest to the highest price.

Aggregated supply and demand curves can be of the following types:

- Stepwise curves containing only step orders (i.e. two consecutive points always have either the same price or the same quantity).
- Linear piecewise and hybrid curves aren't relevant for our model (see [2]).

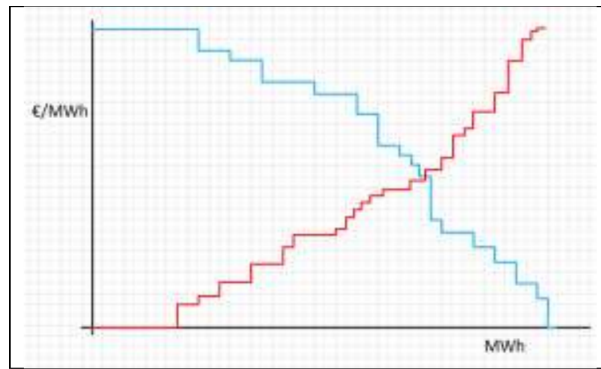


Figure 3: Stepwise aggregated curve [2]

5.2 Complex Orders

A complex order is a set of simple supply stepwise hourly orders (which are referred to as hourly sub-orders) belonging to a single market participant, spreading out along different periods and are subject to a complex condition that affects the set of hourly sub-orders as a whole.

Complex conditions are of two types: Minimum Income Condition (with or without scheduled stop), and Load Gradient.

5.2.1 Minimum Income Condition (MIC)

The Minimum Income economical constraint means that the amount of money collected by the order in all periods must cover its production costs, which is defined by a fix term (representing the startup cost of a power plant) and a variable term multiplied by the total assigned energy (representing the operation cost per MWh of a power plant)

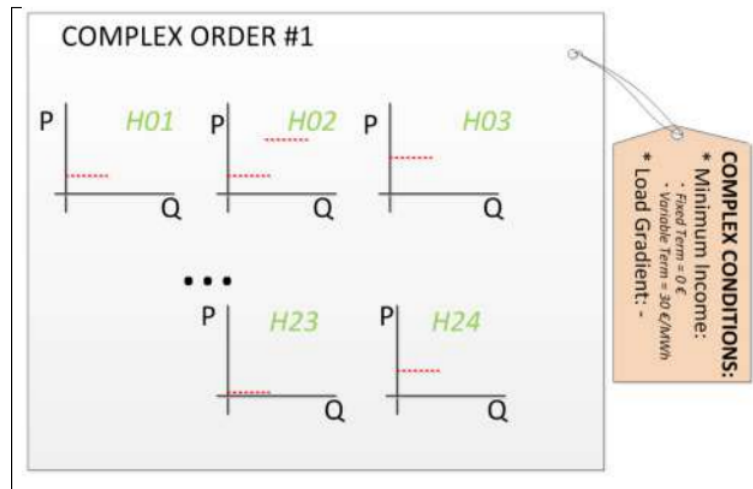


Figure 4: A complex order is composed of a set of hourly sub-orders (indotted line) associated with complex conditions [2]

5.2.2 Scheduled Stop

For the avoidance of abruptly stopping because of MIC, there is the possibility to define a “scheduled stop”. Using a schedule stop will after the deactivation of the MIC: the deactivation will not imply the automatic rejection of all the hourly sub-orders. On the contrary, the first (i.e. the cheapest) hourly sub-order in the periods that contain scheduled stop (up to period 3) will not be rejected but will be treated as any hourly order.

5.2.3 Load Gradient

The Load Gradient constraint means that the amount of energy that is matched by the hourly sub-orders belonging to a Load Gradient order in one period is limited by the amount of energy that was matched by the hourly sub-orders in the previous period. There is a maximum increment / decrement allowed (the same value for all periods). Period 1 is not constrained by the energy matched in the last hour of the previous day. If only one of these values is defined, the other value (i.e. empty) is considered as unconstrained.

5.3 Block Orders

A block order is defined by:

- sense (supply or demand)
e.g. supply
- price limit (minimum price for supply block orders and maximum price for demand block orders)
e.g. 40€/MWh
- number of periods
e.g. Intervals: Hours (3-7), hours (8-19) and hours (22-24)
- volume that can be different for every period
e.g. 80 MWh in the first interval, 220 MWh in the second one, and 40 MWh in the third one

- minimum acceptance ratio
e.g. 0.5

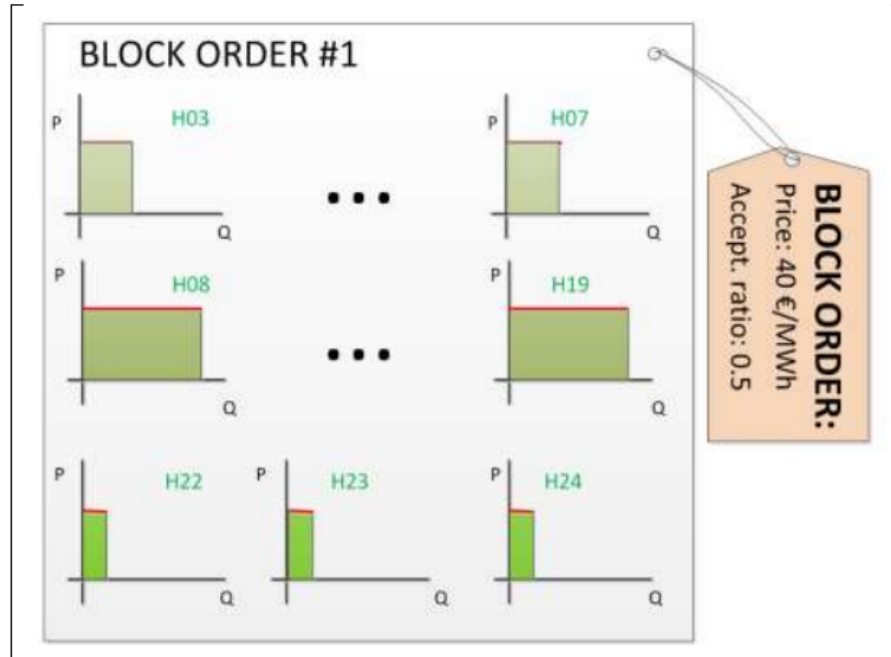


Figure 5: Block order example [2]

In the simplest case, a block order is defined for a consecutive set of periods with the same volume and with a minimum acceptance ratio of 1. These are usually called regular (fill-or-kill) block orders. In general, the periods of the block orders can be non-consecutive, the volume can differ over the periods and the minimum acceptance ratio can be less than 1 (Curtailable Block Orders –partial acceptance is allowed).

5.3.1 Linked Block Orders

Block orders can be linked together, i.e. the acceptance of individual block orders can be made dependent on the acceptance of other block orders. The block which acceptance depends on the acceptance of another block is called “child block”, whereas the block which conditions the acceptance of other blocks is called “parent block”.

5.3.2 Block Orders in an Exclusive Group

An Exclusive group is a set of block orders for which the sum of the accepted ratios cannot exceed 1. In the particular case of blocks that have a minimum acceptance ratio of 1 it means that at most one of the blocks of the exclusive group can be accepted. Between the different valid combinations of accepted blocks the algorithm chooses the one which maximizes the optimization criterion (social welfare).

5.3.3 Flexible Hourly Orders

A flexible “hourly” order is a block order with a fixed price limit, a fixed volume, minimum acceptance ratio of 1, with duration of 1 hour. The hour is not defined by the participant but will be determined by the

algorithm (hence the name “flexible”). The hour in which the flexible hourly order is accepted, is calculated by the algorithm and determined by the optimization criterion.

6 List of Abbreviations

ATC Available Transfer Capacity

CFD Contract for Difference

DAM Day-Ahead Market

ENTSO-E European Network of Transmission System Operators for Electricity

EUPHEMIA Pan-European Hybrid Electricity Market Integration Algorithm

FB Flow Based (model)

FT Fix Term

FTR Financial Transmission Right

LTA Long Term Allocation

MC Market Coupling

MCP Market Clearing Price

MIC Minimum Income Condition

NEMO Nominated Electricity Market Operator

NTH Net Transfer Capacity Handling

PCR Price Coupling of Regions

PRB fully rejected

PPRB partially accepted

PTDF Power Transfer Distribution Factor

PUN Purchase Price Unification

PX Power Exchange

RAM Remaining Available Margin (FB)

SA Schedule Action

SCO Scalable Complex Order

Social welfare consumer surplus + producer surplus + congestion rent across the regions

TSO Transmission System Operator

References

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